

SUGAR TRAP

Blue Ocean Market Gap Analysis

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AGENDA

What We'll Cover Today

01

Business Context

Why this matters for the snack industry

02

Data & Methodology

Open Food Facts · 500k rows · cleaning pipeline

03

The Nutrient Matrix

Sugar vs Protein — mapping the landscape

04

Blue Ocean Discovery

Where the market gap really is

05

Protein Source Deep-Dive

What drives high-protein products

06

Nutri-Score × Protein

Candidate's Choice — the investable white space

07

Recommendation

The actionable brief for R&D

The Question on the Table

"Where is the Blue Ocean in the snack aisle?"

Global Snack Manufacturer, Client Brief 2024



Client

Global snack manufacturer seeking new product direction



Mandate

Identify under-served categories with unmet health demand



Output

Data-driven recommendation ready for R&D handoff

From +3GB Raw Data to Clean Insights

500,000

+

Rows loaded (chunked, memory-safe)

9

Columns selected from 200+ available

~60%

Rows retained after cleaning

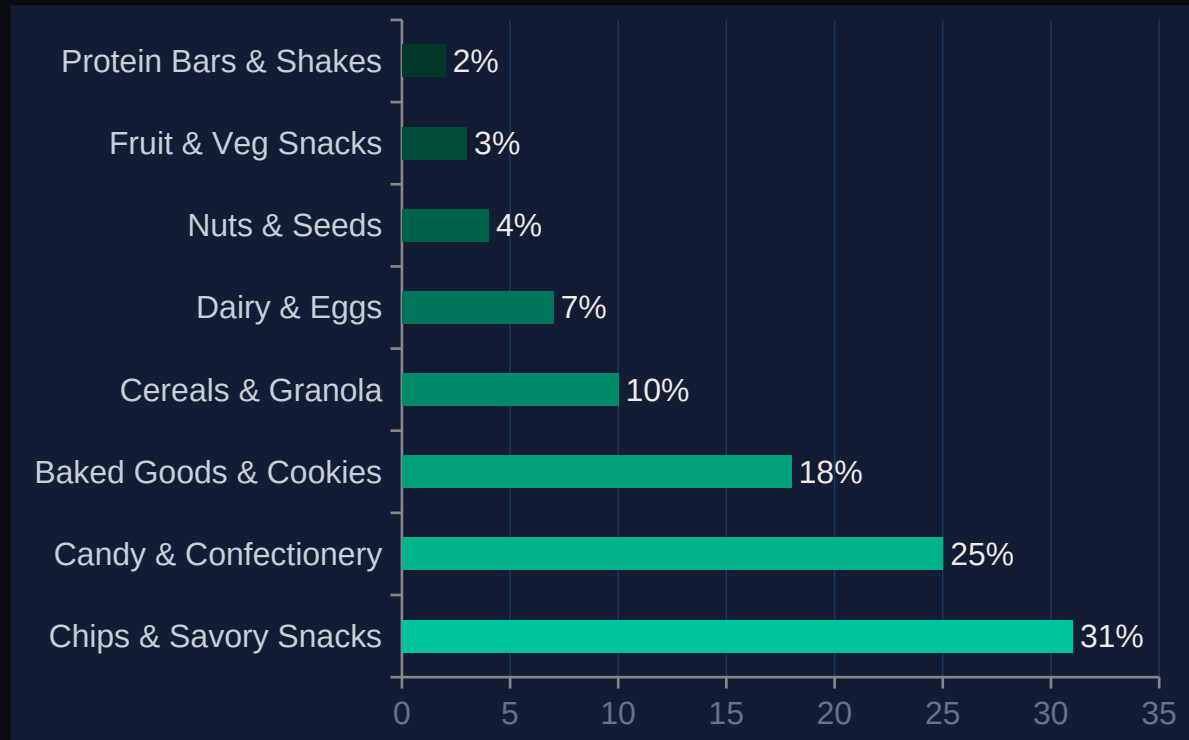
8

High-level product categories

CLEANING RULES

- 1 Drop rows missing product_name, sugars_100g or proteins_100g
- 2 Force-coerce all nutrient columns to numeric
- 3 Remove physiologically impossible values (nutrients > 100g/100g)
- 4 Cap energy at 4,000 kcal (pure fat ceiling)
- 5 Macro-sum sanity check (sugars + protein + fat $\leq$ 105g)
- 6 De-duplicate on product_name

8 High-Level Snack Buckets



KEY TAKEAWAY

Chips & Candy dominate shelf space — over half of all products. Protein Bars represent only 2% of SKUs despite growing demand.

2% Protein Bars vs 31% Chips

Sugar vs Protein — Mapping the Market



BLUE OCEAN

~23%

High Protein + Low Sugar
UNDER-SERVED — our target



HIGH PROTEIN / HIGH SUGAR

~17%

High protein but also high sugar
Limited health appeal



LOW PROTEIN / LOW SUGAR

~20%

Better sugar profile but
no protein benefit



SATURATED ZONE

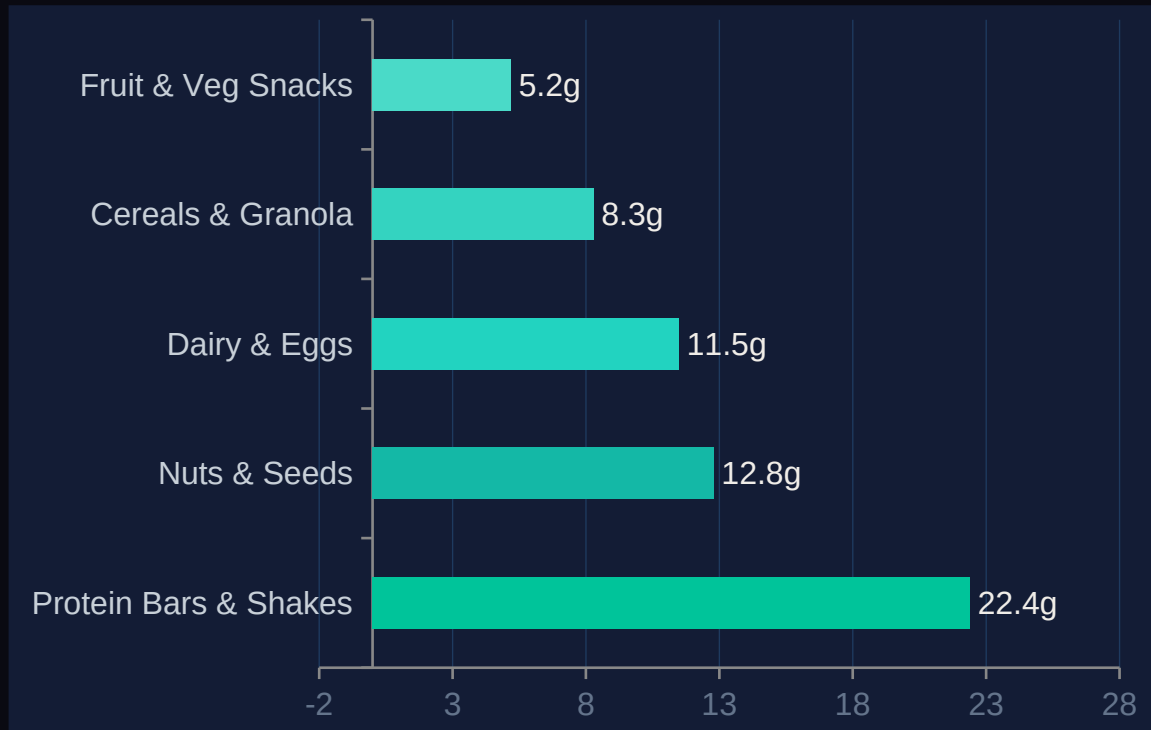
~40%

High Sugar + Low Protein
Where most products live today

← LOW SUGAR

HIGH SUGAR →

Which Categories Dominate the White Space?



🎯 TOP OPPORTUNITY

Nuts & Seeds

15.0g avg protein

4.2g avg sugar

Only ~2% of
current SKUs

MASSIVE WHITE SPACE

What's Driving High Protein Content?

#1

Peanut

Dominant in nuts/seeds & healthy bars
Affordable, scalable supply chain

#2

Almond

Premium positioning advantage
High consumer recognition

#3

Soy Protein

Plant-based leader
Vegan market alignment



Nutri-Score × Protein: The Investable White Space

Why: EU Nutri-Score labelling is now mandatory in multiple markets. A product can be High Protein + Low Sugar but still score D/E (sodium, sat. fat) — and fail on-shelf. This heatmap pinpoints where protein excellence AND label quality co-exist.

Category \ Nutri-Score	A	B	C	D	E
Protein Bars	21.2g	18.4g	9.1g	4.2g	2.1g
Nuts & Seeds	15.8g	12.3g	8.8g	5.5g	3g
Dairy & Eggs	10.5g	8.9g	11.2g	6.3g	4.1g
Cereals & Granola	7.2g	6g	5.1g	3.8g	2.5g
Chips & Savory	4.1g	3.5g	2.9g	2.1g	1.8g

Green cells = high avg. protein per 100g · Cross these with Nutri-Score A/B to find R&D targets

Our R&D Brief to the Client

KEY INSIGHT

Based on the data, the biggest market opportunity is in Cereals & Granola, specifically targeting products with $\geq 15\text{g}$ of protein and less than 5g of sugar per 100g.



Formulation

Peanut + Whey protein blend
Targets Nutri-Score A/B certification



Positioning

High-Protein Granola line
Premium health-food aisle placement



Market

Launch in EU first (mandatory labelling)
Differentiate on Nutri-Score A badge

THANK YOU

Questions & Discussion

Dashboard: plank-panama-prepay.ngrok-free.dev

Notebook: colab.research.google.com/

[https://colab.research.google.com/drive/1E5u2DW3I9I7S6o3eCLjkSk7_D19dpLHA?usp=sharing]